

Method of Application of Propulsion Formulae

Required Outcomes

The main required outcomes of the application of these formulae are:

- To achieve a high enough initial thrust for lift-off.
- To obtain the ideal thrust curve for the given rocket.
- To maintain sufficient chamber pressure to generate thrust without over pressuring the casing.

Steps to Apply the Given Formulae

1. Select an initial chamber pressure from the pressure range and use Equation 1 to obtain the initial K_n value. Different rockets have different initial thrust requirements, so there is no way to accurately predict the initial chamber pressure P_1 . Continued experimentation and research will give more clarity on your choice.
1. Calculate the thrust coefficient C_f from Equation 2, taking P_1 as the initial chamber pressure chosen in Step 1.
2. Calculate the approximate throat area from Equation 3, taking F as an initial approximation value. Initial thrust required by a rocket depends on its mass and launch velocity off the launch pad. For this purpose, initial thrust F can be taken as 6 times the mass of the rocket designed in OpenRocket (generally accepted thrust-to-weight ratio of high-powered rockets at launch is 8:1).
3. Using $Area_{throat}$ and K_n , approximate the dimensions for core diameter d and core length L according to your needs. Keep in mind that these parameters have their own unique effects on the thrust curve of the motor.
4. To get the initial approximation of the thrust curve in OpenMotor, use these initial approximations:
 - Propellant grain type: BATES, KNDX propellant
 - Core diameter d and core length L : choose these values in accordance with Equation 4, according to the requirements of the motor.
 - Grain diameter: at least 2 times the core diameter (increase it if more burn time or impulse is required).
 - Throat diameter: calculated from throat area in Equation 3.
 - Exit diameter: for the initial approximation, take expansion ratio as 8:1 ($8 \times Area_{throat} = Area_{exit}$).
 - Nozzle efficiency: 1.0.
 - Nozzle half angles: research optimal nozzle angles and use the findings as convergent and divergent half angles.

Run the simulation.

5. After achieving the initial approximation of the thrust curve, use the average pressure given by the software (at the bottom of the interface) to calculate the optimal expansion ratio of the nozzle and change the exit diameter accordingly. This will change the thrust curve of the motor, so play around with the other parameters — grain diameter, core diameter, throat diameter, core length, etc. — to achieve the final thrust curve.

Points to Note

- Finding the ideal thrust curve is an iterative process — the greater the number of adjustments and tweaks made, the greater the quality of the thrust curve.
- **Your peak pressure SHOULD NOT exceed 10 MPa.**
- Your calculated values should not deviate from the prescribed range.
- A Python/MATLAB code for these formulas will make the job easier and can allow closer approximations for the first iteration.
- **ONLY USE THE UNITS MENTIONED BELOW THE FORMULAS. DO NOT USE ANY OTHER UNIT.**
- 1 MPa = 1 N/mm².